

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International Advanced Subsidiary and Advanced Level

MARK SCHEME for the October/November 2014 series

9709 MATHEMATICS

9709/13

Paper 1, maximum raw mark 75

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2014 series for most Cambridge IGCSE[®], Cambridge International A and AS Level components and some Cambridge O Level components.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol ∇ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only – often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1	A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through ✓" marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR–2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA –1	This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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1 $(15 \text{ or } {}^{16}C_2) \times 2^4 \times (ax)^2, (20 \text{ or } {}^6C_3) \times 2^3 \times (ax)^3$ $a = \frac{15 \times 2^4}{20 \times 2^3} = \frac{3}{2}$	B1B1 M1A1 [4]	$240a = 160a$ is M0
2 (i) $CB \text{ or } AB = \frac{3}{\tan \frac{\pi}{6}} \text{ or } 3 \tan \frac{\pi}{3}$ Arc or $AC = 3 \times \left[\frac{2\pi}{3} \text{ or } \frac{\pi}{3} \right] \quad (= 2\pi \text{ or } \pi)$ Perimeter $= 6\sqrt{3} + 2\pi$ oe (ii) Area $OABC \quad (2) \times \frac{1}{2} \times 3 \times \text{their } AB$ $ (= 9\sqrt{3} \text{ or } \frac{9\sqrt{3}}{2})$ Area $OADC \quad \frac{1}{2} \times 3^2 \times \left(\frac{2\pi}{2} \text{ or } \frac{\pi}{3} \right) \quad \left(= 3\pi \text{ or } \frac{3\pi}{2} \right)$ Shaded area $9\sqrt{3} - 3\pi$ oe	B1 B1 B1 [3] B1^h B1 B1 [3]	Allow throughout for e.g. $3\sqrt{3}, \sqrt{27}, \sqrt{3^3}, (\sqrt{3})^3, \frac{9}{\sqrt{3}}$ After B0B0 SCB1 for 16.7 Their AB in form $k\sqrt{3}$ After B0B0 SCB1 for 6.16 or 6.17. Allow $(\sqrt{3})^5 - 3\pi$
3 (i) $(3x - 2)^2 + 1$ (ii) $f'(x) = 9x^2 - 12x + 5$ $= \text{their } (3x - 2)^2 + 1$ > 0 (or ≥ 1) hence an increasing function	B1B1B1 [3] B1 M1 A1 [3]	For either of 1st 2 marks bracket must be in the form $(ax + b)^2$ except for SCB2 for $9\left(x - \frac{2}{3}\right)^2 + 1$ Ft from (i). Some reference/recognition Allow > 1. Allow <i>their</i> 1 provided positive. Allow a complete alt method (2/2 or 0/2)
4 (i) $S_P = \frac{2}{1 - \frac{1}{2}}, S_P = \frac{3}{1 - \frac{1}{3}}$ $S_P = 4, S_Q = \frac{9}{2}$ $S_R = 5$ cao (ii) $\frac{4}{1 - r} = \text{their } S_R$ $r = \frac{1}{5}$	M1 A1 A1 [3] M1 A1	At least one correct At least one correct

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$R = 4 + \frac{4}{5} + \frac{4}{25} = 4\frac{24}{25}$ or 4.96 cao	A1 [3]	
5 (i) $(s^2 - c^2)(s^2 + c^2)$ OR $s^2(1 - c^2) - c^2(1 - s^2)$ $\sin^2 \theta - \cos^2 \theta$ $2\sin^2 \theta - 1$ www AG (ii) $2\sin^2 \theta - 1 = \frac{1}{2} \Rightarrow \sin \theta = (\pm)\frac{\sqrt{3}}{2}$ or $(\pm)0.866$ $\theta = 60^\circ$ $\theta = 120^\circ$ $\theta = 240^\circ, 300^\circ$	M1 A1 A1 [3] B1 B1 B1 [4]	OR $\sin^4 \theta - (1 - \sin^2 \theta)^2$ $\sin^4 \theta - (1 - 2\sin^2 \theta + \sin^4 \theta)$ $= 2\sin^2 \theta - 1$ AG OR $\cos 2\theta = -\frac{1}{2} \rightarrow 2\theta = 120, 240$ etc. Ft for $180 - \text{their } 60$ Ft for $180 + \text{their } 60, 360 - \text{their } 60$ Allow $\frac{\pi}{3}, \frac{2\pi}{3}$ etc. Extra sols in range -1
6 (i) $m = \frac{3a + 9 - (2a - 1)}{2a + 4 - a} = \frac{a + 10}{a + 4}$ oe e.g. $\frac{-a - 10}{-a - 4}$ Gradient of perpendicular $= \frac{-(a + 4)}{a + 10}$ oe but not $\frac{-1}{\left(\frac{a + 10}{a + 4}\right)}$ (ii) $(\sqrt{})[(a + 4)^2 + (a + 10)^2] = (\sqrt{})260$ $(\sqrt{})[(a + 4)^2 + (a + 10)^2]$ cao $(2)(a^2 + 14a - 72) (= 0)$ $a = 4$ or -18 cao	M1A1 A1 [3] M1 A1 A1 [4]	cao Allow omission of brackets for M1 Do not ISW. Max penalty for erroneous cancellation 1 mark Allow <i>their</i> $(a + 4), (a + 10)$ from (i). Allow $(-a - 4)^2$ etc. Allow omission of brackets

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7 (i) $OA \cdot OB = -7 + 3 - 3p + p^2$ $(p+1)(p-4) = 0$ $p = -1$ or 4 (ii) $49 + (1 - p^2) + p^2 = 2(1 + 9 + p^2)$ $p = 15$ (iii) $AB = -8\mathbf{i} + 6\mathbf{j}$ Divide AB by $AB = \sqrt{(-8)^2 + 6^2} = 10$ soi Unit vector $= \frac{1}{10}(-8\mathbf{i} + 6\mathbf{j})$ oe cao	M1 DM1 A1 [3] M1 A1 [2] B1 M1 A1 [3]	Correct method for scalar product Equate to zero & attempt to factorise/solve ‘= 0’ implied by answers Scalar result required $p = 15$ used – treat as MR $\rightarrow \frac{1}{\sqrt{353}} \begin{pmatrix} -8 \\ -17 \\ 0 \end{pmatrix}$
8 (i) Minimum since $f''(3) (= 4/3) > 0$ www (ii) $f'(x) = -18x^{-2} (+c)$ $0 = -2 + c$ $c = 2 \rightarrow f'(x) = -18x^{-2} + 2$ $f(x) = 18x^{-1} + 2x (+k)$ $7 = 6 + 6 + k$ $k = -5 \rightarrow f(x) = 18x^{-1} + 2x - 5$ cao	B1 [1] B1 M1 A1 B1[✓] B1[✓] M1 A1 [7]	Sub $f'(3) = 0$. (dep c present) $c = 2$ sufficient at this stage Allow cx at this stage Sub $f(3) = 3$ (k present & numeric (or no) c)
9 (i) $x - 3\sqrt{x} + 2$ or $k^2 - 3k + 2$ or $(3\sqrt{x})^2 = (x+2)^2$ $\sqrt{x} = 1$ or 2 or $k = 1$ or 2 or $x^2 - 5x + 4 (= 0)$ $x = 1$ or 4 $y = 3$ or 6	M1 A1 A1 A1 [4]	OR attempt to eliminate x eg sub $x = \frac{y^2}{9}$ $y^2 - 9y + 18 = 0$ $y = 3$ or 6 $x = 1$ or 4

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(ii) $\int 3x^{\frac{1}{2}} dx - \left[\int (x+2) dx \text{ or attempt at trapezium} \right]$ $2x^{\frac{3}{2}} - \left[\left(\frac{1}{2}x^2 + 2x \right) \text{ or } \frac{1}{2}(y_2 + y_1)(x_2 - x_1) \right]$ $(16 - 2) - \left[\left((8+8) - \left(\frac{1}{2} + 2 \right) \right) \text{ or } \textit{their} \frac{1}{2} \times 9 \times 3 \right]$ $\frac{1}{2}$ OR $\left[\int (y-2) dy \text{ or attempt at trap} \right] - \int \frac{y^2}{9} dy$ $\left[\frac{1}{2}y^2 - 2y \text{ or } \frac{1}{2}(x_1 + x_2)(y_2 - y_1) \right] - \frac{y^3}{27}$ $\left[(18 - 12) - \left(4\frac{1}{2} - 6 \right) \text{ or } \frac{1}{2} \times 5 \times 3 \right] - [8 - 1]$ $\frac{1}{2}$	M1DM1 A1A1 DM1 A1 [6] M1DM1 A1A1 DM1 A1	Attempt to integrate. Subtract at some stage Where $(x_1, y_1), (x_2, y_2)$ is <i>their</i> (1, 3), (4, 6) Apply <i>their</i> 1→4 limits correctly to curve For A mark allow reverse subtn→ $-\frac{1}{2} \rightarrow \frac{1}{2}$ but not reversed limits Apply <i>their</i> 3→6 limits correctly to curve
10 (a) (i) $(a+b)^{\frac{1}{3}} = 2, (9a+b)^{\frac{2}{3}} = 16$ $a+b=8, 9a+b=64$ $a=7, b=1$ (ii) $x = (7y+1)^{\frac{1}{3}}$ (x/y interchange as first or last step) $x^3 = 7y+1$ or $y^3 = 7x+1$ $f^{-1}(x) = \frac{1}{7}(x^3 - 1)$ cao Domain of f^{-1} is $x \geq 1$ cao (b) $\frac{dy}{dx} = \left[\frac{1}{3}(7x^2 + 1)^{-\frac{2}{3}} \right] \times [14x]$ When $x=3, \frac{dy}{dx} = \frac{1}{3} \times (64)^{\frac{2}{3}} \times 42 \quad \left(= \frac{7}{8} \right)$ $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = \frac{7}{8} \times 8$ 7	B1B1 M1 A1 [4] B1 B1 B1 B1 [4] B1B1 M1 DM1 A1 [5]	Ignore 2nd soln (−9, 17) throughout Cube etc. & attempt to solve Correct answers without any working 0/4 ft on from <i>their</i> a, b or in terms of a, b ft on from <i>their</i> a, b or in terms of a, b A function of x required Accept $>$. Must be x Use chain rule